Fruit Mill



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Fruit mill

What it is used for

A fruit mill crushes or grinds prepared fruit into smaller particles or a coarse mash for further processing. Correct preparation, machine adjustment, and hygienic operation help produce a consistent feed while reducing contamination and equipment damage.

Main parts

- Fruit feed hopper
- Grinding or crushing rotor, blades, or rollers
- Screen or sizing plate where fitted
- Motor and drive assembly
- Discharge outlet and collection container
- Protective guards, control switch, and emergency stop

Before you start

1. Confirm the fruit type, required particle size, batch quantity, and intended downstream process.
2. Inspect the hopper, grinding parts, screen, discharge outlet, guards, and drive components for damage.
3. Ensure all food-contact surfaces are clean, sanitised, and suitable for food processing.
4. Check that the collection container is clean, food-grade, and correctly positioned under the outlet.
5. Remove stones, metal fragments, damaged fruit, and other foreign material; wash and trim the fruit according to the process.
6. Verify that guards, emergency stop, electrical connections, and other safety controls are functioning.

How to use

1. Switch off and isolate power before opening the mill, changing screens, or adjusting grinding parts.
2. Fit the approved screen or grinding configuration for the required product consistency.
3. Close all covers and confirm that guards are securely fitted before starting.
4. Position the collection container and check that the discharge path is clear.
5. Switch on the mill and allow it to reach normal operating speed before feeding fruit, as directed by the manufacturer.
6. Feed prepared fruit gradually through the hopper without exceeding the machine's rated capacity.

7. Use only the approved feed mechanism or pusher; never insert hands or tools into the hopper while the mill is running.
8. Monitor the discharge consistency, machine load, vibration, and sound during processing.
9. Reduce the feed rate or stop the mill if the outlet blocks, the motor overloads, or abnormal vibration occurs.
10. Stop the machine and isolate power before removing a blockage or checking internal parts.
11. Collect the processed fruit in clean containers and transfer it promptly to the next processing stage.
12. Inspect the product for consistency and any signs of foreign material or contamination.
13. At the end of the batch, switch off and isolate the machine before removing remaining fruit residue.
14. Disassemble only designated removable parts and clean and sanitise them according to the approved procedure.

⚠ Safety

- Only trained personnel should operate or clean the fruit mill.
- Never bypass guards or reach into the hopper, rotor, blades, or rollers while the machine is running.
- Isolate power before clearing jams, changing screens, or servicing the mill.
- Do not exceed the machine's rated capacity or use it for materials not approved by the manufacturer.
- Keep hands, clothing, and utensils away from moving parts and pinch points.
- Use suitable PPE and follow safe handling procedures when working with sharp blades or heavy removable parts.
- Prevent water from entering electrical components unless the equipment is specifically designed for the approved washdown method.

Care & storage

- Clean and sanitise the hopper, grinding parts, screen, and discharge outlet after use.
- Inspect blades, rollers, screens, seals, and fasteners for wear or damage.
- Check motor ventilation, drive components, and emergency-stop function regularly.
- Keep the machine dry and protect electrical components during cleaning.
- Arrange servicing if particle size becomes inconsistent, the motor overheats, or vibration increases.

Common mistakes to avoid

- Feeding fruit with stones or other foreign material still present.
- Overloading the hopper and causing blockages or motor strain.
- Opening the mill or clearing a blockage while it is energised.
- Using unclean containers or failing to sanitise food-contact parts.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.